# NASCAR SPONSORSHIP VISIBILITY

Exploratory Data Analysis Report

2024 Cup Series · 5 Sponsors · 36 Races · 180 Data Points

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Project Week 4 · EDA & Correlation Analysis

## Key Findings at a Glance

| | |
|---|---|
| **8 / 8** | Correlations statistically significant (all $p < 0.0001$) |
| **r = 0.871** | Strongest predictor: Laps Led vs Reddit Mentions |
| **10.75×** | Reddit uplift: Top-5 finishes vs P11+ finishes |
| **9.04×** | Visibility multiplier: Race wins vs non-wins |
| **Key surprise** | Laps Led outperforms Finish Position on every channel |

## 1. Executive Summary

This report presents the Exploratory Data Analysis (EDA) for the NY Racing NASCAR Sponsorship Visibility project. It covers 180 sponsor-race observations (36 races × 5 sponsors) across the 2024 NASCAR Cup Series season. The analysis tested eight performance-to-visibility relationships across three media channels — Reddit, YouTube, and news coverage — and produced eight statistically significant correlations (all $p < 0.0001$).

**Primary Finding:**
Laps Led is the single strongest predictor of sponsor visibility across every channel tested ($r = 0.871$ for Reddit, 0.864 for News, 0.854 for YouTube). This directly challenges the initial assumption that Finish Position would be the primary performance driver. The implication for the Week 5 model is significant: laps led should carry the highest individual weight, not finish position.

**Sponsors Analysed:**

| Sponsor | Driver | Car # | Team | Tier | Wins | Avg Finish | Laps Led |
|---|---|---|---|---|---|---|---|
| NAPA Auto Parts | Chase Elliott | #9 | Hendrick Motorsports | Top | 4 | 10.0 | 770 |
| Monster Energy | Tyler Reddick | #45 | 23XI Racing | Top | 4 | 9.2 | 641 |
| Fastenal | Chris Buescher | #17 | RFK Racing | Mid | 1 | 16.0 | 144 |
| Dollar Tree | J.H. Nemechek | #42 | Legacy Motor Club | Lower | 0 | 27.6 | 0 |
| Love's Travel Stops | Todd Gilliland | #34 | Front Row Motorsports | Lower | 0 | 25.7 | 0 |

## 2. Descriptive Statistics

The tables below summarise race performance and exposure metrics for each sponsor across the 36-race 2024 Cup Series season.

### 2.1 Race Performance Statistics

| Sponsor | Mean Finish | Median | Std Dev | Best | Worst | Wins | Top-5 % |
|---|---|---|---|---|---|---|---|
| NAPA Auto Parts | 10.0 | 9.5 | 6.98 | 1 | 25 | 4 | 33.3% |
| Monster Energy | 9.2 | 7.5 | 7.69 | 1 | 29 | 4 | 38.9% |
| Fastenal | 16.0 | 15.5 | 7.73 | 1 | 30 | 1 | 11.1% |
| Love's Travel Stops | 25.7 | 26.0 | 6.41 | 12 | 36 | 0 | 0.0% |
| Dollar Tree | 27.6 | 28.0 | 5.73 | 15 | 40 | 0 | 0.0% |

### 2.2 Exposure / Visibility Statistics

| Sponsor | Total Reddit | Avg Reddit | Total YT Views | Avg YT Views | Total News | Avg News |
|---|---|---|---|---|---|---|
| NAPA Auto Parts | 3,922 | 108.9 | 13,685,016 | 380,139 | 594.7 | 16.5 |
| Monster Energy | 3,716 | 103.2 | 13,028,714 | 361,909 | 498.4 | 13.8 |
| Fastenal | 786 | 21.8 | 2,442,193 | 67,839 | 108.0 | 3.0 |
| Love's Travel Stops | 145 | 4.0 | 436,847 | 12,135 | 22.0 | 0.6 |
| Dollar Tree | 193 | 5.4 | 480,730 | 13,354 | 24.5 | 0.7 |

Top-tier sponsors (NAPA, Monster Energy) generate 18-27x more Reddit mentions and YouTube views than lower-tier sponsors on a per-race average basis. This tier gap underscores the importance of distinguishing performance-driven visibility from baseline team-reputation visibility when building the scoring model.

## 3. Correlation Analysis

Pearson correlation coefficients and two-sided p-values were calculated for all eight key performance-to-visibility pairs. The full results are shown below, ranked by absolute correlation strength.

### 3.1 Correlation Results Table

| # | Variable 1 | Variable 2 | r | $R^2$ | P-Value | Strength | Sig. |
|---|---|---|---|---|---|---|---|
| 1 | Laps Led | Reddit Mentions | +0.871 | 0.759 | < 0.0001 | Strong | *** |
| 2 | Laps Led | News Mentions | +0.864 | 0.746 | < 0.0001 | Strong | *** |
| 3 | Laps Led | YouTube Views | +0.854 | 0.729 | < 0.0001 | Strong | *** |
| 4 | Laps Led | YouTube View Share | +0.762 | 0.581 | < 0.0001 | Strong | *** |
| 5 | Finish Position | YT View Share | -0.699 | 0.489 | < 0.0001 | Mod-Strong | *** |
| 6 | Finish Position | YouTube Views | -0.615 | 0.378 | < 0.0001 | Moderate | *** |
| 7 | Finish Position | News Mentions | -0.585 | 0.342 | < 0.0001 | Moderate | *** |
| 8 | Finish Position | Reddit Mentions | -0.579 | 0.335 | < 0.0001 | Moderate | *** |

*** $p < 0.01$ — Highly significant. All 8 correlations exceed the $p < 0.05$ threshold.

### 3.2 Sponsor-Level Correlation Breakdown

Correlations were also computed within each sponsor to check for heterogeneity. Dollar Tree is the notable outlier — it shows no significant finish-to-Reddit correlation ($r = -0.145$, $p = 0.400$), suggesting its low brand awareness means results do not register meaningfully with the r/NASCAR community.

| Sponsor | Finish vs Reddit (r) | p | Finish vs YouTube (r) | p | Finish vs News (r) | p |
|---|---|---|---|---|---|---|
| NAPA Auto Parts | -0.524 | 0.001 | -0.555 | 0.000 | -0.540 | 0.001 |
| Monster Energy | -0.578 | 0.000 | -0.635 | 0.000 | -0.657 | 0.000 |
| Fastenal | -0.541 | 0.001 | -0.590 | 0.000 | -0.500 | 0.002 |
| Love's Travel Stops | -0.496 | 0.002 | -0.365 | 0.029 | -0.345 | 0.039 |
| Dollar Tree | -0.145 | 0.400 | -0.253 | 0.136 | -0.399 | 0.016 |

## 4. Visualizations

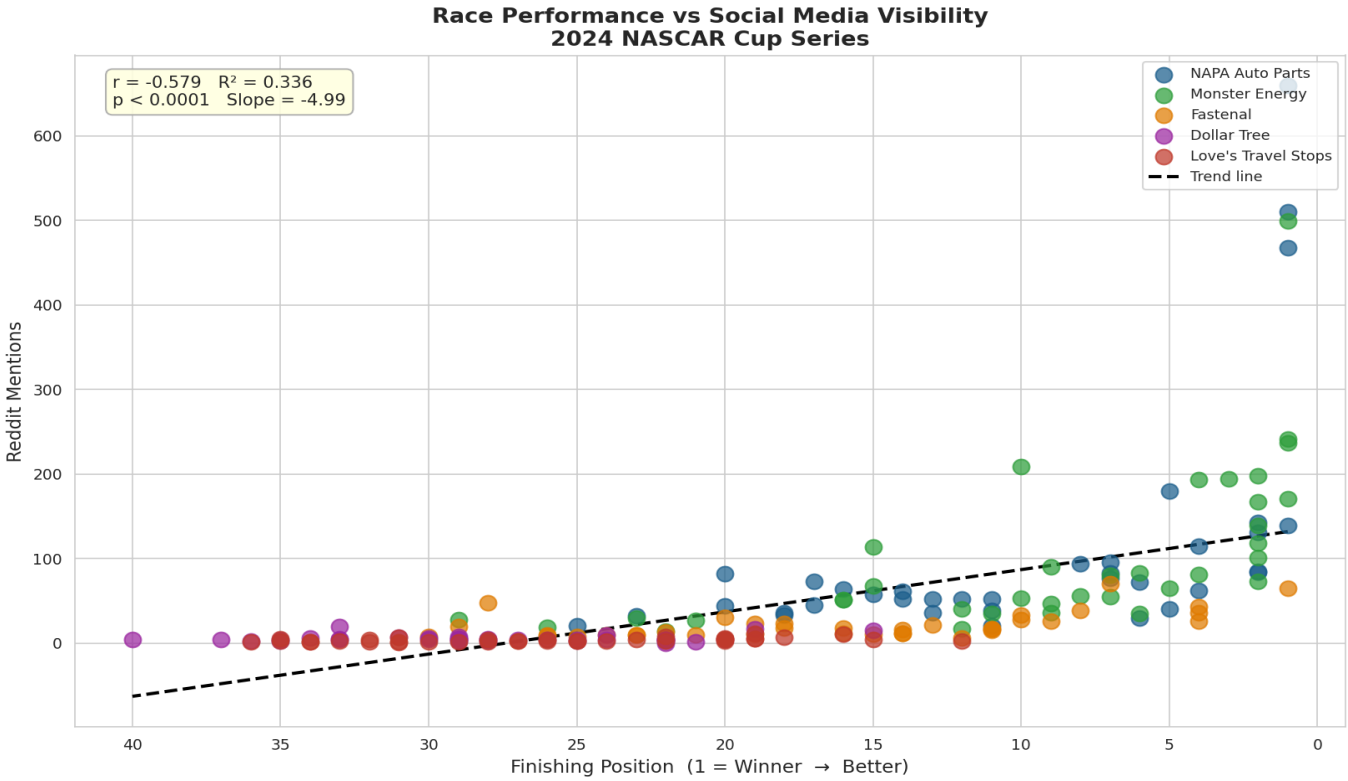


*Figure 1 — Primary Scatter: Race Finishing Position vs Reddit Mentions. Trend line shows negative relationship (better finishes → more mentions). r = -0.579, p < 0.0001.*

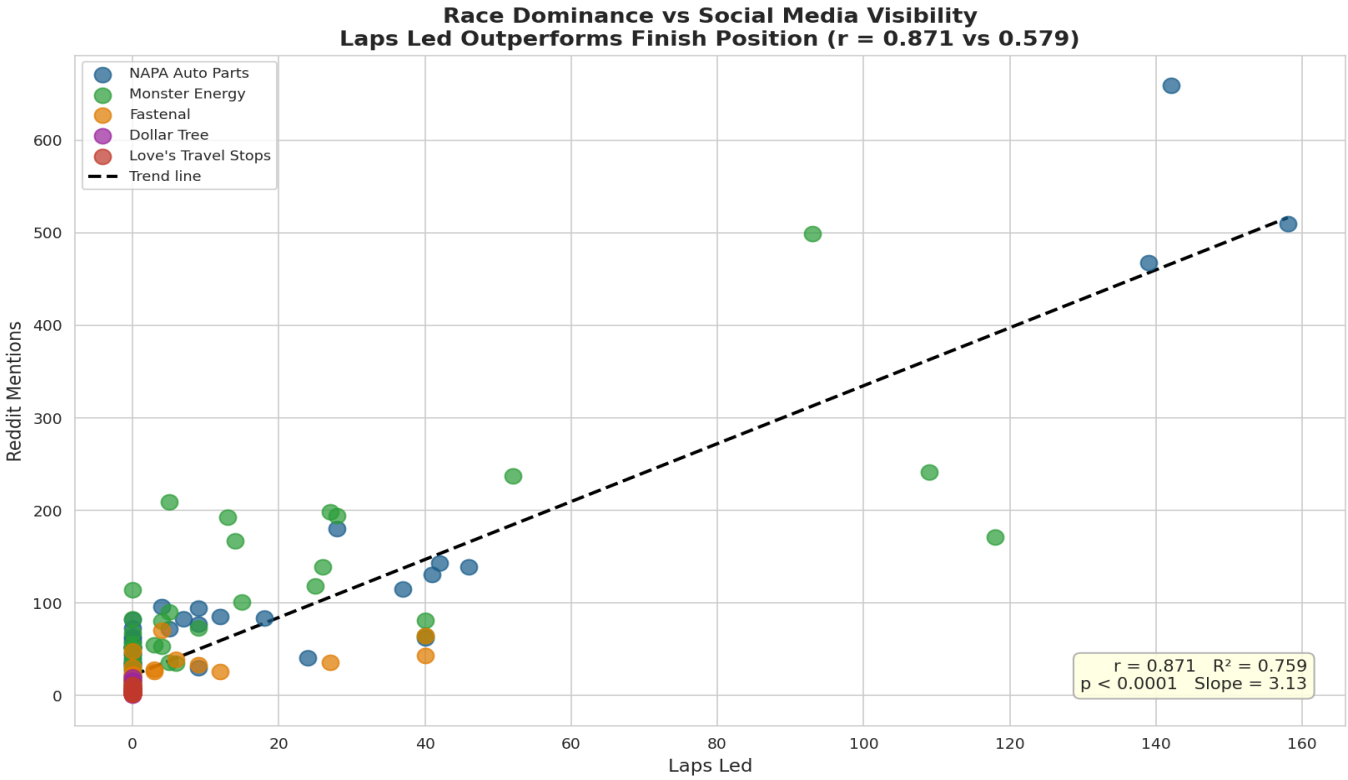


*Figure 2 — Race Dominance vs Reddit: Laps Led vs Reddit Mentions. Stronger relationship (r = +0.871) than Finish Position — the key surprise finding.*

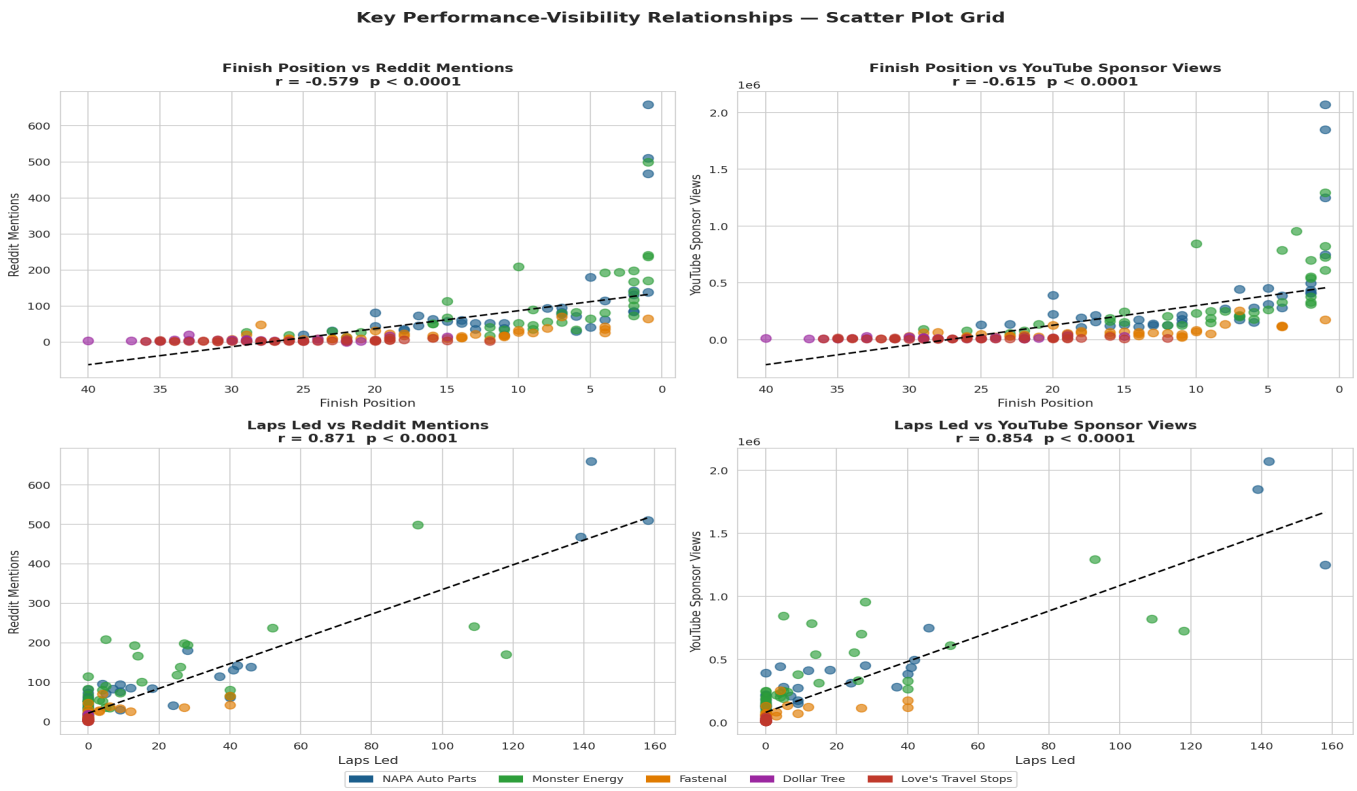


*Figure 3 — 2×2 Scatter Grid: All four key performance-exposure pairs with trend lines. Laps Led relationships (bottom row) consistently steeper than Finish Position (top row).*

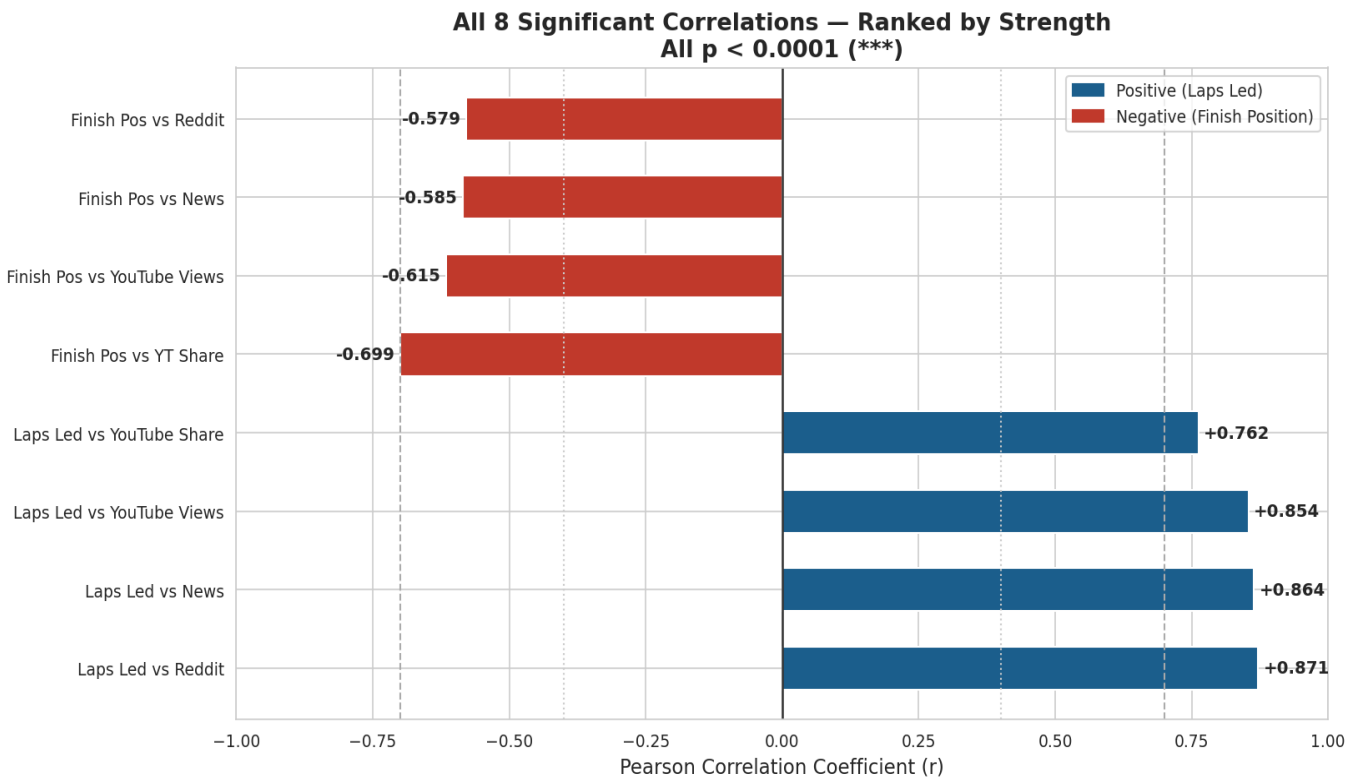


*Figure 4 — Correlation Summary: All 8 significant correlations ranked by strength. Laps Led (blue) dominates; Finish Position (red) still significant but weaker.*

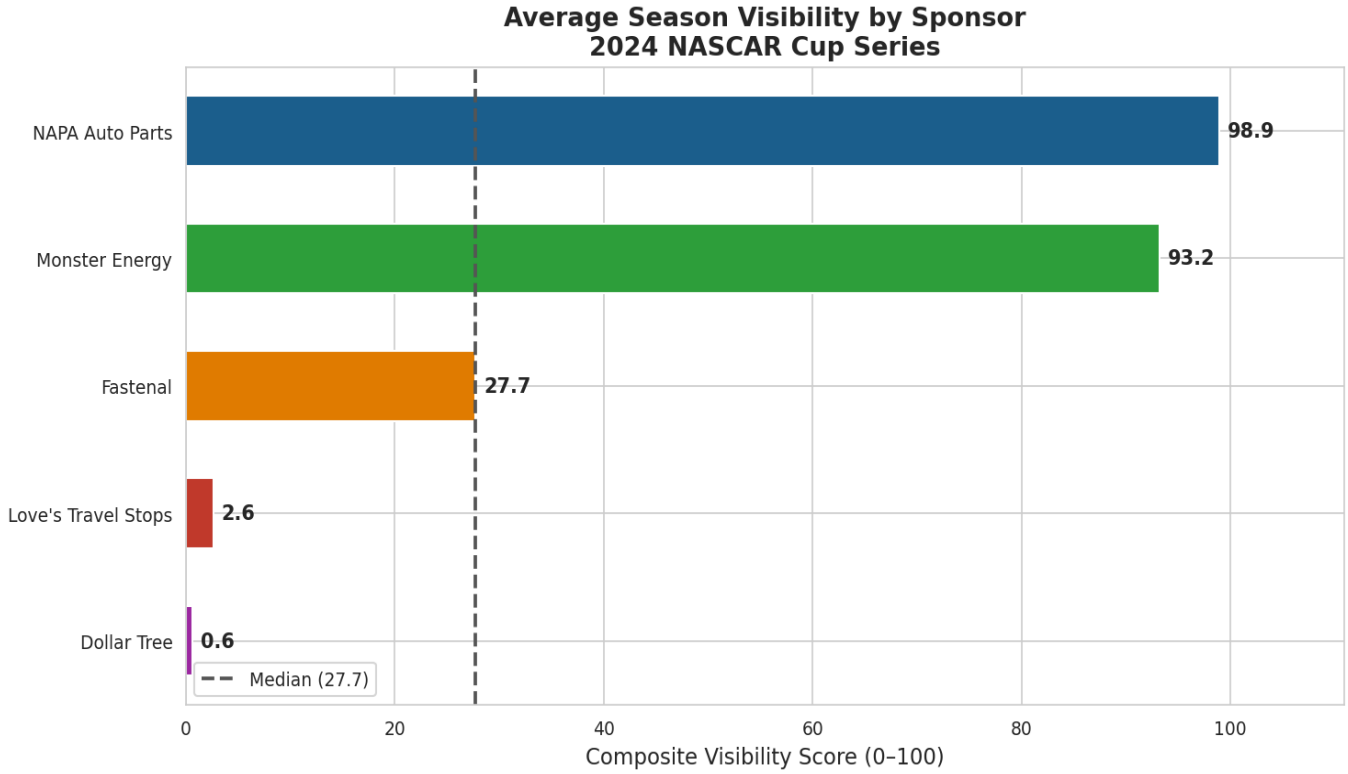


*Figure 5 — Composite Visibility Score by Sponsor. NAPA and Monster Energy lead by a wide margin; tier gap is stark.*

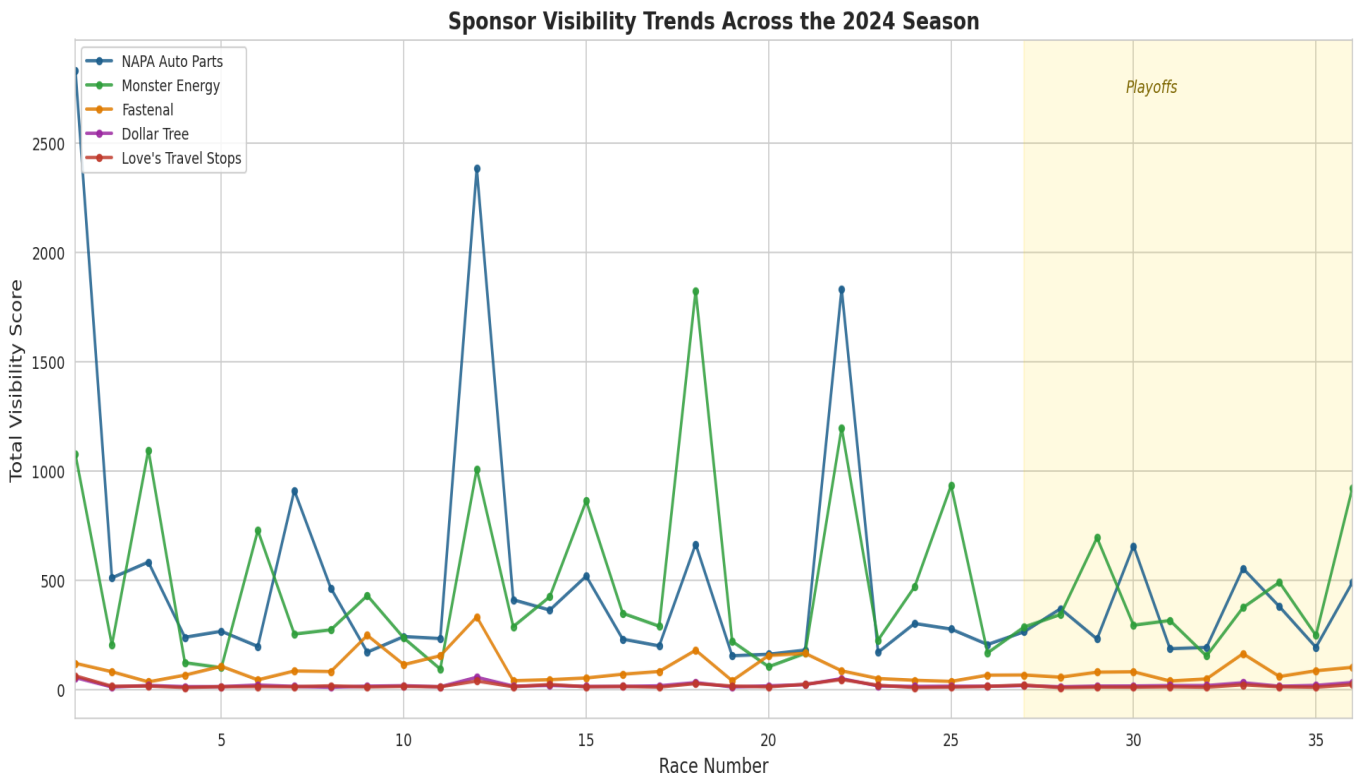


*Figure 6 — Season Visibility Trends. NAPA and Monster Energy dominate throughout; playoff shading (Race 27+) shows no clear amplification.*

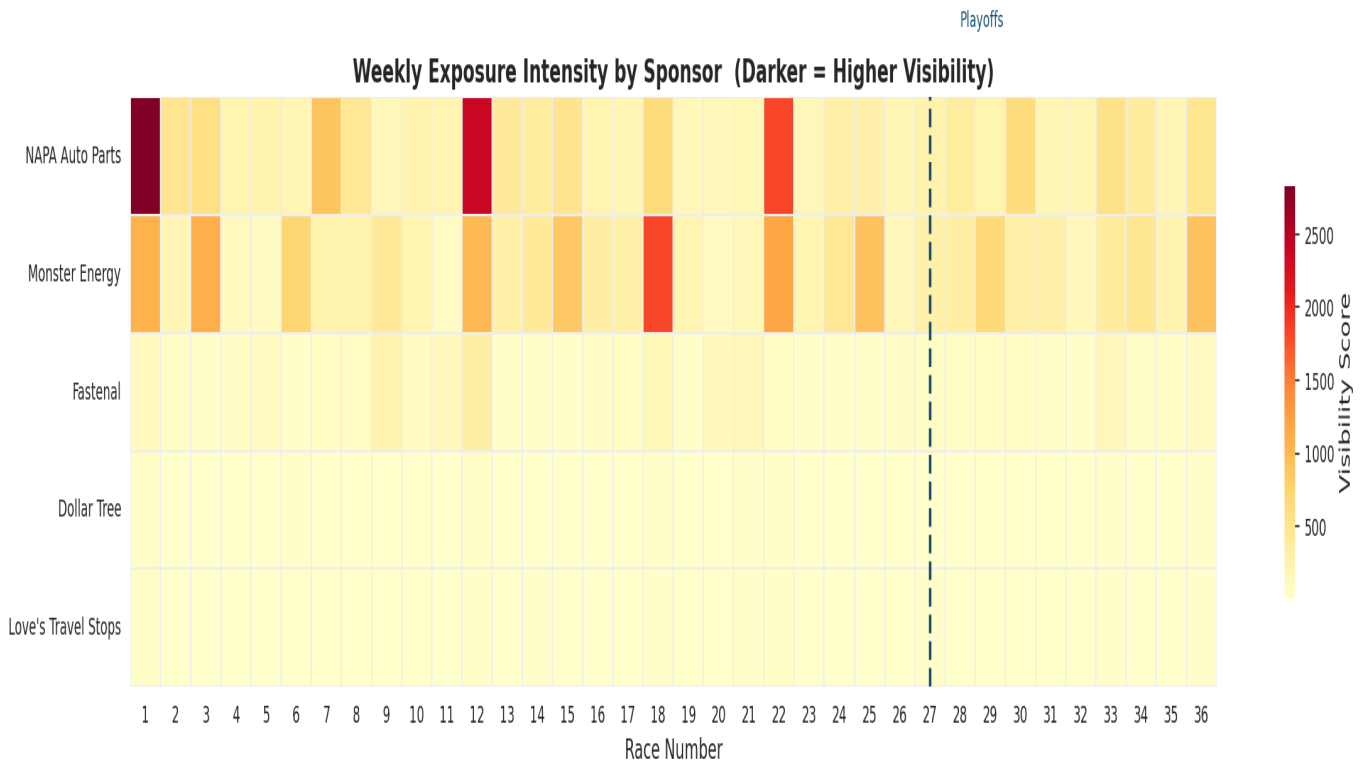


*Figure 7 — Weekly Exposure Intensity Heatmap. Win races (bright cells) are clearly visible for top-tier sponsors. Lower-tier sponsors remain dim throughout.*

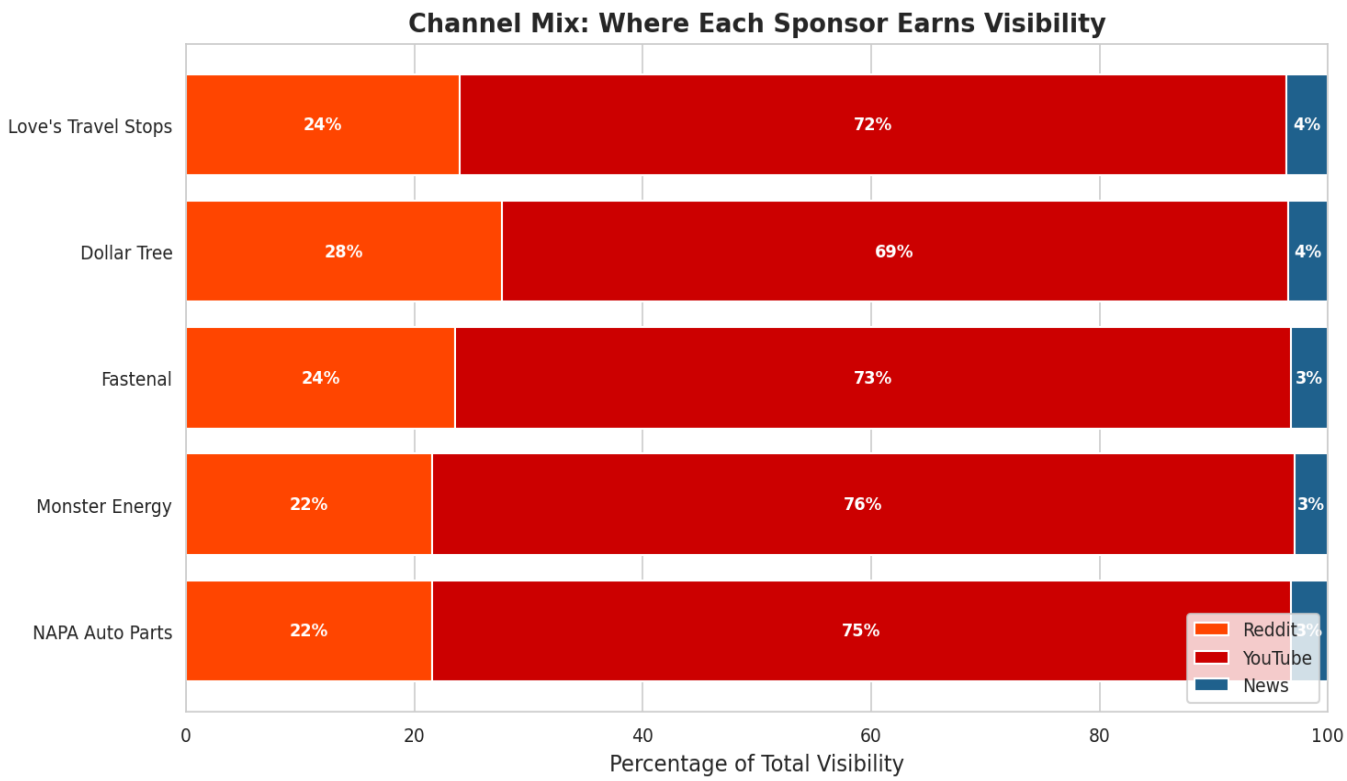


*Figure 8 — Channel Mix by Sponsor. YouTube dominates raw volume for all sponsors; Reddit carries a higher proportional share for lower-tier.*

## 5. Testable Hypotheses for the Scoring Model

The following five hypotheses emerge directly from the EDA findings. Each is stated in falsifiable form and ranked by model development priority.

### H1 (Priority 1)

*Laps Led is the primary performance driver of cross-channel sponsor visibility*

| | |
|---|---|
| **Evidence** | r = 0.871 (Reddit), 0.864 (News), 0.854 (YouTube) — all strong, all $p < 0.0001$ |
| **Comparison** | Laps Led outperforms Finish Position by 0.275 points average across channels |
| **Model implication** | Weight laps led at 40-45% of composite score. Finish position retains secondary role (20-25%). |
| **Test method** | Week 5 regression: compare model R-squared with and without laps-led weight |

### H2 (Priority 1)

*Race wins generate disproportionate visibility — a non-linear "win bonus" exists*

| | |
|---|---|
| **Evidence** | Win-to-non-win total visibility ratio: 9.04x. This far exceeds a linear position effect. |
| **Comparison** | Top-5 vs P11+ Reddit ratio: 10.75x. Winning amplifies visibility non-linearly. |
| **Model implication** | Add a discrete win-bonus multiplier (~3-4x) on top of position weight. |
| **Test method** | Compare linear vs piecewise regression; test significance of win indicator variable. |

### H3 (Priority 2)

*Playoff races (Race 27-36) do not amplify visibility — regular season may perform equally or better*

| | |
|---|---|
| **Evidence** | Playoff average visibility: 174.5. Regular season average: 241.2. Playoff boost: -27.7% (lower, not higher). |
| **Possible explanation** | Playoff races may feature more competitive grids with fewer laps-led opportunities for any single driver. |
| **Model implication** | Do not apply a blanket playoff multiplier. Consider a race-prestige tier instead. |
| **Test method** | Two-sample t-test: playoff vs regular season mean visibility. Check race-prestige variable. |

| H4 (Priority 2) | |
|---|---|
| *Reddit and News are more performance-sensitive channels than YouTube* | |
| **Evidence** | Laps Led: Reddit r=0.871, News r=0.864, YouTube r=0.854 — all strong but Reddit/News marginally higher. |
| **Channel structure** | YouTube views reflect broader race viewership; sponsor-specific views track performance more tightly. |
| **Model implication** | Maintain provisional weights: Reddit 30%, News 25%, YouTube 20%, with engagement adding 25%. |
| **Test method** | Compare channel-level R-squared values in the Week 5 regression model. |

| H5 (Priority 3) | |
|---|---|
| *Dollar Tree and Love's require a minimum-visibility floor adjustment — performance signals are weak* | |
| **Evidence** | Dollar Tree finish/Reddit correlation: $r = -0.145$, $p = 0.400$ — not statistically significant. |
| **Contrast** | All other sponsors show $p < 0.040$ or better for the same relationship. |
| **Model implication** | Flag lower-tier sponsors for a "brand-awareness offset" — their visibility floor is driven by brand recognition, not race performance. |
| **Test method** | Compare residuals from performance-only model for each sponsor tier. |

## 6. Recommendations for the Scoring Model

Based on EDA findings, the following structure is recommended for the Week 5–6 visibility scoring model:

| Component | Metric | Recommended Weight | Rationale |
|---|---|---|---|
| Primary driver | Laps Led | 40-45% | Strongest cross-channel predictor (r = 0.854-0.871) |
| Secondary driver | Finish Position | 20-25% | Significant but weaker than laps led; retain as secondary |
| Win bonus | Binary win indicator | ×3-4 multiplier | 9x visibility uplift on win races is non-linear |
| Social channel | Reddit mention volume | 30% of exposure | Most performance-sensitive channel; provisional weight |
| Engagement | Reddit upvotes + comments | 25% of exposure | Amplifies raw mention signal with quality weight |
| News channel | Weighted news mentions | 25% of exposure | Strong laps-led signal (r=0.864); tier-weighted source |
| Video channel | YouTube sponsor views | 20% of exposure | Reliable but slightly lower than Reddit/News sensitivity |
| Tier offset | Sponsor tier baseline | TBD in Week 6 | Lower-tier sponsors need floor adjustment (H5) |

**Pending Sarah sign-off:** Model weights are provisional pending VP of Partnerships approval before Week 6 implementation.

## 7. Methodology

| | |
|---|---|
| **Analysis period** | 2024 NASCAR Cup Series — 36 races (February–November 2024) |
| **Dataset** | 180 sponsor-race observations: 5 sponsors × 36 races, 23 columns |
| **Correlation method** | Pearson correlation (r) with two-sided p-value via scipy.stats.pearsonr |
| **Significance threshold** | $p < 0.05$ (standard); all 8 pairs exceed $p < 0.0001$ |
| **Outlier treatment** | Retained — outliers represent real events (wins, flagship races) relevant to model |
| **Visualization tools** | matplotlib 3.x, seaborn; charts exported at 150 DPI |
| **Data sources** | Race results: Racing Reference / manual; Reddit: PRAW API; YouTube: Data API v3; News: manual Google search |
| **Known limitation** | Reddit/YouTube metrics reflect values at collection time, not race time |